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Pin assignments based on bandwidth

Abstract

In example implementations, an apparatus is provided. The apparatus includes a cable interface, a video interface, an integrated chip, and a controller. The cable interface is to connect a host device via a cable. The video interface is to output video signals corresponding to visual data. The controller is communicatively coupled to the cable interface, the video interface, and the integrated chip. The controller assigns pairs of pins of the cable to transmit video signals based on the bandwidth used by the video signals. The controller also causes the host device to re-assign the pairs of pins of the cable to transmit the video signals.

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Background/Summary

BACKGROUND

(1) Universal serial bus (USB) cables can be used to transfer data between devices. USB cables have evolved over the years to become faster and provide more functionality. For example, USB cables have improved from the original USB cable to USB 2.0, USB 3.0 and now USB type-C (USB-C) cables.

(2) USB-C cables are 24 pin double-sided connectors that can be used to transfer data and provide power. The USB-C cables can be used to transfer data at rates of 10 gigabytes (GB) per second (GB/s) to 20 GB/s.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a block diagram of an example system that includes an apparatus to automatically assign pins of a high-speed cable based on an amount of video bandwidth used by a video output of the present disclosure;
- (2) FIG. 2 is a block diagram of an example of the apparatus of the present disclosure;
- (3) FIG. 3 is a more detailed block diagram of the example apparatus of the present disclosure;
- (4) FIG. 4 is a block diagram of example pins of the high-speed cable of the present disclosure;
- (5) FIG. 5 is a flow chart of an example method for automatically assigning pins of a high-speed cable based on an amount of video bandwidth used by a video output of the present disclosure; and
- (6) FIG. 6 is a block diagram of an example non-transitory computer-storage medium storing instructions executed by a processor to automatically assign pins.

DETAILED DESCRIPTION

- (7) Examples described herein provide an apparatus and method for automatically changing pin assignments in a USB-C cable based on an amount of video bandwidth used by a display. As discussed above, USB-C cables are 24 pin double-sided connectors that can be used to transfer data and provide power. The USB-C cables can be used to transfer data at rates of 10 gigabytes (GB) per second (GB/s) to 20 GB/s.
- (8) The configuration of the USB-C cables can be manually changed by changes made in a basic input/output system (BIOS) of a host device. For example, a user can go into the BIOS of a computer to change the pin assignments for a USB-C cable. This may be a complicated and laborious process for some users.
- (9) The present disclosure provides an apparatus (e.g., a docking station or a display) that can automatically change the USB-C pin assignments based on an amount of video bandwidth used by a display. For example, for some resolution settings, a monitor may use a certain amount of video bandwidth. The USB-C default pin assignment may use 2 of the four available pairs for assignment for video and 2 of the four available pairs for assignment for USB 3.0 data transfer. However, for some images at certain resolutions, 2 pairs of pins in the USB-C cable may not provide enough bandwidth.
- (10) In an example, the apparatus calculates the amount of video bandwidth that is to be used when a display is initially connected. Based on the amount of video bandwidth, the apparatus may automatically assign the appropriate pins in the USB-C cable to transfer video data. The apparatus may then automatically cause the host device to change settings in the host device that correspond to the pin assignments of the USB-C cable set in the apparatus. As a result, the pin assignments of the USB-C cable may be automatically switched.
- (11) FIG. 1 illustrates an example system **100** of the present disclosure. In an example, the system **100** may include an apparatus **102**, a display **104**, and a host device **106**. The apparatus **102** may be communicatively coupled to the display **104** and the host device **106**. In an example, the host device **106** may be communicatively coupled to the apparatus **102** via a high-speed cable **108**.
- (12) In an example, the high-speed cable **108** is a cable that can transmit data at rates of greater than 5 gigabits (Gb) per second (s). In an example, the high-speed cable can transmit data at rates between 5 Gb/s to 40 Gb/s. The high-speed cable **108** can also transfer video data and provide power to the host device **106**. The high-speed cable **108** includes pins that can be reconfigured to perform different functions or transfer different types of data. An example of the high-speed cable **108** capable of the above features is a USB-C cable. However, it should be noted that any type of video/data cable or connection that can be re-configured and provide high-speed data transfer may be within the scope of the present disclosure.
- (13) In an example, the apparatus **102** may be a docking station or may be combined with the

display **104** as a single unit. The apparatus **102** may include an interface **112** to connect to the high-speed cable **108**, a controller **110**, and an interface **114** to connect to the display **104**. The controller **110** may be a single processor, or a combination of processors or devices that operate together, to automatically assign or re-assign pins of the high-speed cable **108** depending on the amount of video bandwidth used by the display **104** or any other video output of the apparatus **102**.

(14) As discussed in further details below, the controller **110** may calculate the amount of video bandwidth that is used and determine if the assignment of the pins of the high-speed cable **108** should be changed. For example, certain pins or pairs of pins in the high-speed cable **108** can be assigned to transmit data or video signals/data. Each assignable pin of the high-speed cable **108** can transmit a certain amount of video data. The controller **110** may determine how many of the assignable pins are assigned to transmit video signals/data to determine an amount of video bandwidth that is available on the high-speed cable **108** under a current pin assignment.

(15) The amount of video bandwidth may be based on characteristics of a video image that is to be transmitted. For example, the amount of video bandwidth may be based on a resolution (e.g., a number of pixels per horizontal line and vertical line), a refresh rate of the image, a number of bits of the video image, and the like. The video bandwidth may indicate how much bandwidth for transmitting video should be available in the high-speed cable **108**.

(16) Based on the calculation, the video bandwidth used by the video image may be greater than the available video bandwidth of the high-speed cable **108** under the current pin assignment. The controller **110** can determine that the pins should be re-assigned and automatically re-assign the pins. The re-assignment of the pins may cause the host device **106** to set a configuration in accordance with the assignment of the pins set by the controller **110**. In other words, the pin assignment may occur under control of the controller **110** and without manually intervention. The host device **106** may then transfer data in accordance with the pin assignments set by the apparatus **102**.

(17) In an example, the host device **106** may be a desktop computer, a laptop computer, a tablet computer, and the like. The host device **106** may want to display video on the display **104**. As noted above, to change the pin assignments of the high-speed cable **108**, a user previously would manually change the assignments via the BIOS of the host device **106**. However, the apparatus **102** of the present disclosure allows the pin assignments to be changed or set automatically.

(18) FIG. 2 illustrates a block diagram of an example of the apparatus **102** of the present disclosure. In an example, the apparatus **102** may include a high-speed cable interface **202**, a controller **204**, an integrated chip (IC) **206**, and a video interface **208**. The high-speed cable interface **202** may be a port or coupling that allows the high-speed cable **108** to be connected to the apparatus **102**. The high-speed cable interface **202** may have pins or openings that correspond to the pins of the high-speed cable **108** to form a secure connection. For example, the pins of the high-speed cable **108** may be inserted into the corresponding openings of the high-speed cable interface **202**.

(19) In an example, the video interface **208** may provide a connection to the display **104** to output signals corresponding to visual data. For example, the visual data may include video, video signals, video data, images, and the like. The video interface **208** may transmit video data received from the host device **106** to the display **104**. In an example, where the apparatus **102** is a docking station, the video interface **208** may be a port or a connection that coincides with a type of video cable used to connect to the display **104**. For example, the video interface **208** may be a display port, a high definition media interface (HDMI) port, a thunderbolt port, a video graphics adapter (VGA) port, and the like. In an example, where the apparatus **102** is part of the display **104**, the video interface **208** may be a connection between the IC **206** and the graphics processor of the display **104**.

(20) In an example, the IC **206** may be a monolithic semiconductor chip that includes a set of electronic circuits. The IC **206** may be printed as a unit using photolithography rather than being constructed one transistor or layer at a time as done with discrete processors. Examples of the IC **206** may include a scaler chip or a display port (DP) multi-streaming transport (MST) hub chip.

The IC **206** may be a scaler chip when the apparatus **102** is combined as part of the display **104** or may be a DP MST hub chip when the apparatus **102** is deployed as a stand-alone docking station.

(21) In an example, the IC **206** may calculate the amount of video bandwidth used by the display **104**. For example, when the display **104** is initially connected to the apparatus **102** via the video interface **208**, the IC **206** may communicate with the display **104** to determine an amount of video bandwidth that is used by the display **104** or video that is output from the video interface **208**. In an example, the amount of video bandwidth that is used may be based on a highest resolution of the display **104** and/or the amount of bandwidth used by video that is output at the highest resolution.

(22) In an example, the IC **206** may determine if the current pin assignments for the high-speed cable **108** are sufficient to provide the amount of video bandwidth used by the video or the display **104**. In other words, the IC **206** may determine if the pin assignments for the high-speed cable **108** provide enough bandwidth or data throughput to transmit the video in a desired resolution, format, refresh rate, and the like, to the display **104**. If the current pin assignments are not sufficient, the IC **206** can control the controller **204** to change or re-assign the pin assignments for the high-speed cable **108**.

(23) For example, the IC **206** may communicate the amount of video bandwidth that is used to the controller **204**. The IC **206** may communicate with the controller **204** over a low-speed bus (e.g., an inter-integrated circuit (I2C) bus). The IC **206** may cause the controller **204** to set the pin assignments in accordance with the amount of video bandwidth that is used by the video or the display **104**.

(24) In an example, the controller **204** may be a processor or an application specific integrated circuit (ASIC). The controller may execute instructions stored in memory to perform the functions described herein or may be programmed (e.g., in the case of an ASIC) to perform a function.

(25) In an example, the controller **204** may then present a single option for the pin assignment of the high-speed cable **108** to the host device **106**. Thus, the controller **204** may cause the host device **106** to automatically set the pin assignments for the high-speed cable **108** to provide enough bandwidth for the video and/or the display **104**. In other words, the pin assignments are not changed manually through the BIOS of the host device **106**.

(26) FIG. 3 illustrates a more detailed block diagram of the apparatus **102**. In an example, the apparatus **102** may include the high-speed cable interface **202**, the controller **204** and the IC **206**, as illustrated in FIG. 2, and discussed above. In an example, the apparatus **102** may include a plurality of video interfaces **208.sub.1** to **208.sub.n** (herein also referred to individually as a video interface **208** or collectively as video interfaces **208**).

(27) In an example, the IC **206** may determine an amount of video bandwidth used by the video and/or displays each time a display **104** is connected to a video interface **208.sub.1** to **208.sub.n**. As a result, the pin re-assignment performed by the controller **204** may be dynamically changed. For example, when a first display is connected to the video interface **208.sub.1**, the resolution may be low enough that the current pin assignments of the high-speed cable **108** are sufficient to transmit the video data to the first display.

(28) However, at a later time after the first display is connected, a second display may be connected to the video interface **208.sub.2**. The IC **206** may determine that the video bandwidth used for the resolution of the second display is higher than the available bandwidth on the current pin assignments of the high-speed cable **108**. As a result, the IC **206** may cause the controller **204** to change the pin assignments and cause the host device **106** to also change the pin assignments accordingly in response to the determination that the video bandwidth used for the second display is higher than the available bandwidth on the current pin assignments of the high-speed cable **108**.

(29) At a later time after the pin assignments for the high-speed cable **108** are implemented for the second display, the second display may be disconnected. The IC **206** may determine the pin assignments and communicate the pin assignments to the controller **204**. The IC **206** can cause the controller **204** to change the pin assignments again to accommodate the lower resolution and lower

video bandwidth use of the first display.

(30) In an example, the pin assignments for the first display and the second display can be stored in memory. As a result, if the second display is connected to the video interface **208.sub.2** again, the IC **206** may obtain the pin assignments to be implemented from the memory. Some processing can be skipped (e.g., the determination of the video bandwidth used for the resolution of the second display) and the pin assignments can be implemented more efficiently.

(31) In an example, the apparatus **102** may include a multiplexer (mux) **310** and a hub **312**. The multiplexer **310** may be a USB-C 4:6 multiplexer when the high-speed cable **108** is a USB-C cable. The mux **310** maps the pin assignments and changes to the pin assignments made by the controller **204**. Thus, the mux **310** routes the data received via the high-speed cable **108** to the hub **312** and/or the IC **206** based on the pin assignments. For example, if the pin assignments assign each of the high-speed data pins of the high-speed cable for video data, the mux **310** may route the video data from each of the high-speed data pins to the IC **206**. However, if some of the high-speed data pins are assigned for video data and other high-speed data pins are assigned for USB 3.0 data transfer, the mux **310** may route the video data to the IC **206** and the USB 3.0 data transfer to the hub **312**.

(32) In an example, the hub **312** may be used to connect other USB 3.0 devices to the apparatus **102**. For example, when data is transferred via USB 3.0 protocol over the high-speed cable **108**, the data may be routed to the hub **312**. The hub **312** may then transmit the data to the appropriate device connected to the apparatus **102**. The hub **312** may include interfaces or input/output ports to connect to the USB 3.0 devices.

(33) FIG. 4 illustrates a block diagram of example pins of a high-speed cable **108** of the present disclosure. The pin layout illustrated in FIG. 4 is an example of pins for a USB-C cable.

(34) In an example, the high-speed cable **108** may include 12 pairs of pins (A1-A12 and B1-B12). Some of the pins may be assigned as ground pins and provide power (e.g., ground (GND) pins (A1, B12) and (A12, B1) and VBUS pins (A4, B9) and (A9, B4)). Some pins may be assigned to provide digital audio (e.g., display port (DP) auxiliary (AUX)– pin B8 and DP AUX+ pin A8). Some pins may be assigned to communicate configuration information (e.g., side band channels OC1 pin A5 and OC2 pin B5). Some pins may be assigned to provide USB 2.0 data transfer rates (e.g., USB D+ pin A6 and USB D– pin A7).

(35) However, the high-speed cable **108** may include 4 pairs of high-speed data pins that can be re-assigned to transfer different types of data. For example, the pin pairs (A2, A3), (B10, B11), (B2, B3), and (A10, A11) can be switched between video data for display port (DP) or USB 3.0 data transfer. Thus, in an example when referring to re-assignment of pins or changing pin assignments, the controller **204** may change the assignment of the pairs of high-speed data pins (A2, A3), (B10, B11), (B2, B3), and (A10, A11). The pairs of high-speed data pins may be re-assigned or changed between transmitting data using USB 3.0 and transmitting video data for DP.

(36) In an example, when the controller **204** presents a single pin configuration for the high-speed cable **108** to the host device **106**, the single pin configuration may be related to the assignment of the pairs of high-speed data pins to either using USB 3.0 or transmitting video data for DP.

(37) Notably, when the pin assignments are changed or re-assigned, the high-speed cable **108** still provides power to the host device **106**. In addition, other types of data may still be transferred using a lower transfer rate of USB 2.0 (e.g., via pins A6 and A7).

(38) FIG. 5 illustrates a flow diagram of an example method **500** for automatically assigning pins of a high-speed cable based on an amount of video bandwidth used by a video output of the present disclosure. In an example, the method **500** may be performed by the apparatus **102** or the apparatus **600** illustrated in FIG. 6, and described below.

(39) At block **502**, the method **500** begins. At block **504**, the method **500** determines an amount of video bandwidth used by a video output. In an example, the amount of video bandwidth may be determined when a display is initially connected to a video interface or communicatively coupled to a processor that is executing the method **500**. For example, a processor in an apparatus (e.g., a

docking station or a part of the display) may communicate with the display to determine how much video bandwidth will be used by the video output.

(40) In an example, the amount of video bandwidth used by the video output may be determined each time a new connection is detected. For example, there may be a plurality of video interfaces, and another display may be connected to the video interface. The amount of video bandwidth used by the video output of the newly connected display and/or a combination of all video outputs may be determined.

(41) In an example, the amount of video bandwidth may be calculated based on a resolution of the display. In an example, the calculation may be based on a highest resolution of the display.

(42) At block **506**, the method **500** determines that the amount of video bandwidth is greater than an available bandwidth of a high-speed cable based on a first pair of pins of the high-speed cable that are assigned to transmit video. In an example, the pins of the high-speed cable may have a default pin configuration. The pin configuration may be assigned by a controller and the assignments may be mapped to the physical pins of the high-speed cable by a multiplexer. For example, two pairs of high-speed pins may be set to transfer data using USB 3.0 and two pairs of the high-speed pins may be set to transfer video data for DP connections. However, the amount of video data that can be transferred by the two pairs of high-speed pins assigned to video data may not be sufficient for the amount of video bandwidth calculated in block **504**. In other words, the amount of video bandwidth used by the video output may be greater than the amount of video data that can be transferred by the default setting of two pairs of high-speed pins for video data.

(43) At block **508**, the method **500** assigns a second pair of pins of the high-speed cable to transmit the video based on the amount of video bandwidth. For example, other pins of the high-speed cable may be able to transmit video data, but may be currently assigned to transmit other types of data.

These other pins can be re-assigned to transmit video data instead of the other types of data. Using a USB-C cable as an example, two pairs of the high-speed pins assigned to transfer data using USB 3.0 can be re-assigned to transmit video data. The change in pin assignments can be mapped by a 4:6 mux within the docking station or display apparatus. For example, two pairs of pins (A2, A3) and (B10, B11) may have a default assignment to transfer data using USB 3.0. The processor may change the assignment for the pairs of pins (A2, A3) and (B10, B11) to transfer video data instead.

(44) At block **510**, the method **500** causes a host device to assign the second pair of pins of the high-speed cable to transmit the video. In an example, a processor or controller of an apparatus that performs the method **500** may communicate with the host device. The processor may present a single configuration option of the pins or pin assignment to the host device. The host device may then be forced to select the single pin assignment presented by the processor. The host device may automatically implement the pin assignment as instructed by the processor.

(45) At block **512**, the method **500** provides the video to the video output via the first pair of pins and the second pair of pins of the high-speed cable. For example, the host device may transmit video data over the pins that are assigned to transmit video data. The video data may then be provided to the video interface or output to be shown on the display in the desired resolution. At block **514**, the method **500** ends.

(46) FIG. **6** illustrates an example of an apparatus **600**. In an example, the apparatus **600** may be the apparatus **102**. In an example, the apparatus **600** may include a processor **602** and a non-transitory computer-readable storage medium **604**. The processor **602** may be a central processing unit (CPU), a graphical processing unit (GPU), an application specific integrated circuit (ASIC), and the like. Although a single processor **602** is illustrated in FIG. **6**, it should be noted that multiple processors **602** may be deployed to execute the instructions stored in the non-transitory computer-readable storage medium **604** in a distributed computing environment.

(47) In an example, the non-transitory computer-readable storage medium **604** may include a hard-disk drive, a random access memory (RAM), a read-only memory (ROM), a solid state drive, and the like. The non-transitory computer-readable storage medium **604** may include instructions **606**,

608, 610, and 612. The processor **602** may access the non-transitory computer-readable storage medium **604** to obtain the instructions **606, 608, 610, and 612.** The processor **602** can then execute the instructions **606, 608, 610, and 612** to perform various functions.

(48) In an example, the instructions **606** may include instructions to cause a processor to receive an indication that an amount of bandwidth used by a video output is greater than an available amount of bandwidth on two pairs of pins of a universal serial bus type-C (USB-C) cable. The instructions **608** may include instructions to cause a host device connected via the USB-C cable to assign, based on the amount of video bandwidth, two additional pairs of pins to transmit video data. The instructions **610** may include instructions to cause the host device to receive the video data over the two pairs of pins and the two additional pairs of pins of the USB-C cable. The instructions **612** may include instructions to cause the host device to transmit the video data to the video output.

(49) It will be appreciated that variants of the above-disclosed and other features and functions, or alternatives thereof, may be combined into many other different systems or applications. Various presently unforeseen or unanticipated alternatives, modifications, variations, or improvements therein may be subsequently made by those skilled in the art which are also intended to be encompassed by the following claims.

Claims

1. An apparatus, comprising: a cable interface to: receive, while connected with a host device via a cable, video data from pins in the cable; a video interface to: transmit, while connected to a display, the video data to the display, and an integrated circuit chip to: determine, in response to detecting a connection between the video interface and the display, an amount of video bandwidth necessary for transmission of the video data to the display by the video interface based on a refresh rate of a video image included within the video data; and a controller to: control, in response to determining the amount of video bandwidth, the host device to reconfigure a current assignment of the pins in the cable based on the determined amount of video bandwidth in response to a determination that the current assignment is insufficient to provide the determined amount of video bandwidth.
2. The apparatus of claim 1, wherein the cable comprises a universal serial bus type-C (USB-C) cable.
3. The apparatus of claim 2, wherein the USB-C cable comprises four pairs of high-speed pins, and the four pairs of high-speed pins are assigned to transmit the video data.
4. The apparatus of claim 1, further comprising: a 4:6 multiplexer to map pairs of the pins that are re-assigned.
5. The apparatus of claim 1, further comprising: the integrated circuit chip to calculate the amount of video bandwidth.
6. The apparatus of claim 5, wherein the integrated circuit chip comprises a display port multi-streaming transport hub chip of a docking station.
7. The apparatus of claim 5, wherein the integrated circuit chip comprises a scaler chip of a display.
8. The apparatus of claim 5, wherein the controller is communicatively coupled to the cable interface, the video interface, and the integrated circuit chip.
9. The apparatus of claim 1, wherein the controller is to determine the amount of video bandwidth based on characteristics of the video image.
10. The apparatus of claim 1, wherein the controller is to determine the amount of video bandwidth based on a number of bits of the video image.
11. The apparatus of claim 1, wherein the controller is to determine the amount of video bandwidth based on a resolution of the display.
12. A method, comprising: determining, by a processor in response to detecting a connection between a video output and a display, an amount of video bandwidth used by the video output; determining, by the processor, that the amount of video bandwidth is greater than an available

bandwidth of a high-speed cable based on a first pair of pins of the high-speed cable that are assigned to transmit video data; assigning, by the processor, a second pair of pins of the high-speed cable to transmit the video data based on the amount of video bandwidth; causing, by the processor, a host device to assign the second pair of pins of the high-speed cable to transmit the video data; and providing, by the processor, the video data to the video output via the first pair of pins and the second pair of pins of the high-speed cable, wherein the determining the amount of video bandwidth comprises: receiving, by the processor, the amount of video bandwidth that is used from an integrated circuit chip in communication with the display, and wherein the integrated circuit chip calculates the amount of bandwidth based on a refresh rate of a video image included within the video data.

13. The method of claim 12, wherein the amount of video bandwidth is calculated when the display is initially connected to the video output.

14. The method of claim 13, wherein the amount of video bandwidth is calculated based on a resolution of the display.

15. The method of claim 12, wherein the causing the host device to assign the second pair of pins comprises: presenting, by the processor, a single configuration option to the host device that includes the first pair of pins and the second pair of pins of the high-speed cable assigned to transmit the video data.

16. The method of claim 12, wherein the integrated circuit chip comprises a scaler chip or a display port media stream transport hub chip.

17. A non-transitory computer-readable storage medium encoded with instructions executable by a processor, the non-transitory computer-readable storage medium comprising: instructions to cause the processor to receive an indication that an amount of bandwidth used by a video output is greater than an available amount of bandwidth on two pairs of pins of a universal serial bus type-C (USB-C) cable; instructions to cause a host device connected via the USB-C cable to assign two additional pairs of pins of the USB-C cable to transmit video data; instructions to cause the host device to receive the video data over the two pairs of pins and the two additional pairs of pins of the USB-C cable; and instructions to cause the host device to transmit the video data to the video output, wherein the instructions to cause the processor to receive the indication are performed each time a new display is communicatively coupled to an integrated circuit chip that is communicatively coupled to the processor, wherein the indication comprises receiving, by the processor, the amount of video bandwidth that is used from the integrated circuit chip in communication with a display, wherein the integrated circuit chip calculates the amount of bandwidth based on a refresh rate of a video image included within the video data.

18. The non-transitory computer-readable storage medium of claim 17, wherein the two additional pairs of pins of the USB-C cable comprise pin pairs (A2, A3), (B10, B11), (B2, B3), and (A10, A11).
